Skill Check 4 Review FRQ Solutions
AP Computer Science A

Name _____ Period_____

1. An environmental club has a supply budget, a number of students, and a price per reusable bottle. Assume budget and bottlePrice are positive and students is greater than 0. Use the variables below in your calculations; do not hard-code the calculated values.

```
int budget = 74;
int students = 12;
double bottlePrice = 3.25;
```

(a) Declare a double constant named `TAX_RATE` with the value `0.06`. Then declare and initialize a double variable named `priceWithTax` that represents the price of one bottle including tax.

(b) Declare and initialize a double variable named `budgetPerStudent` that represents the budget divided equally among the students. Preserve the fractional part using a cast, and then print the value.

(c) Declare and initialize an int variable named `wholeBottles` that represents the number of whole bottles the club can buy at `priceWithTax`. Use a cast to discard the fractional part, and then print the value.

(d) A student writes the following statement for part (b).

```
double budgetPerStudent = (double)(budget / students);
```

State the value stored in `budgetPerStudent`, explain why the statement does not preserve the fractional part, and write a corrected statement.

(a)
```
final double TAX_RATE = 0.06;
double priceWithTax = bottlePrice * (1 + TAX_RATE);
```
priceWithTax stores 3.445.

(b)
```
double budgetPerStudent = (double) budget / students;
System.out.println(budgetPerStudent);
```
Output: 6.166666666666667

(c)
```
int wholeBottles = (int) (budget / priceWithTax);
System.out.println(wholeBottles);
```
Output: 21

(d)
```
double budgetPerStudent = (double) budget / students;
```
The original stores 6.0 because budget / students performs integer division (74 / 12 = 6) before the cast. Cast an operand before dividing, as shown.

© The learn Project

Name _____ Period_____

2. A race organizer has a container of water, a number of runners, and a cup capacity. Assume totalOunces and cupCapacity are positive and runners is greater than 0. Use the variables below in your calculations; do not hard-code the calculated values.

```
int totalOunces = 275;
int runners = 24;
double cupCapacity = 12.5;
```

(a) Declare a double constant named `SPILL_RATE` with the value `0.05`. Then declare and initialize a double variable named `usableOunces` that represents the amount of water remaining after 5% is lost.

(b) Declare and initialize a double variable named `ouncesPerRunner` that represents the original water divided equally among the runners. Preserve the fractional part using a cast, and then print the value.

(c) Declare and initialize an int variable named `fullCups` that represents the number of completely full cups that can be poured from `usableOunces`. Use a cast to discard the fractional part, and then print the value.

(d) A student writes the following statement for part (c).

```
int fullCups = usableOunces / cupCapacity;
```

Explain why the statement does not compile, identify the data type produced by the division, and write a corrected statement.

(a)
```
final double SPILL_RATE = 0.05;
double usableOunces = totalOunces * (1 - SPILL_RATE);
```
usableOunces stores 261.25.

(b)
```
double ouncesPerRunner = (double) totalOunces / runners;
System.out.println(ouncesPerRunner);
```
Output: 11.458333333333334

(c)
```
int fullCups = (int) (usableOunces / cupCapacity);
System.out.println(fullCups);
```
Output: 20

(d)
```
int fullCups = (int) (usableOunces / cupCapacity);
```
The division produces a double. Java does not implicitly narrow a double to an int, so the original line does not compile. The explicit cast above discards the fractional part.

© The learn Project